# Formal Methods in Software Engineering

## Re-introduction to Logic, Spring 2026

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# Why Formal Methods?

# Motivation

Before we define verification formally, here is *why* it matters.

Software bugs have caused deaths, $billion losses, and mission failures — even when tests passed.

**Ariane 5 (1996)** — Overflow in 64⟶16-bit conversion destroyed a $370M rocket 37 s after launch. Reused code from Ariane 4 *without re-verification*.

**Therac-25 (1985–87)** — Race condition in radiation machine caused massive overdoses. At least 3 deaths.

**Intel FDIV (1994)** — Pentium division error in rare cases. $475M recall; discovered by a mathematician, not by Intel's tests.

**Knight Capital (2012)** — Faulty trading software deployment. Lost $440M in 45 minutes. Bankrupt within days.

In each case, the system's behavior was never *proven* to match its specification.

**Formal methods** turn correctness into a *mathematical question* that machines can help answer. Instead of checking *some* executions, we reason about *all* of them.

# Formal Reasoning: A Taste

Here is the *kind* of reasoning formal methods make precise and machine-checkable.

**Scenario:** A server has three properties documented in its specification:

1. If authentication succeeds, a session is created.
2. If a session is created, the user can access the resource.
3. Authentication succeeded.

In logical notation, with propositions $A$ = "auth succeeds", $S$ = "session created", $R$ = "resource accessible":

$$\underbrace{A \to S}_{\text{premise 1}} \quad \underbrace{S \to R}_{\text{premise 2}} \quad \underbrace{A}_{\text{premise 3}}$$

**Derivation:** From $A$ and $A \to S$ we get $S$ (modus ponens). From $S$ and $S \to R$ we get $R$ (MP again).
**Conclusion:** the user can access the resource.

This is a *proof* — a finite chain of justified steps from premises to a conclusion. Formal methods scale this idea to entire programs: premises are code + pre-conditions, and we prove that post-conditions follow.

# What Is Verification?

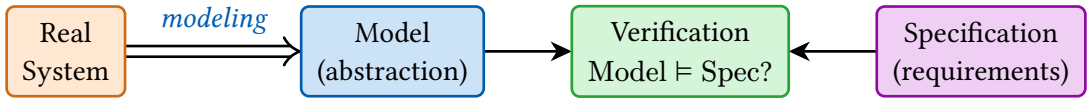
Software verification is the process of establishing that a system *meets its specification*. Three ingredients are needed:

**Definition 1**: A *model* is a mathematical representation of a system — a finite-state machine, a logical formula, a program abstraction. It captures *what the system does* (or can do), abstracting away irrelevant details.

**Definition 2**: A *specification* is a precise, formal statement of *desired behavior*: a pre-condition/post-condition pair, an invariant, a temporal property. It captures *what the system should do*.

**Definition 3**: *Verification* is checking whether a model satisfies a specification: Model $\vDash$ Spec.

# What Is Verification? [2]




*Example*: Consider a function `abs(x)` that should return $|x|$.
**Model:** the program code (or its logical encoding).
**Specification:** result $\geq 0$ and (result $= x \lor$ result $= -x$).
**Verification:** prove that for *all* inputs $x$, the model satisfies the specification.

**Formal methods** = build a model + write a specification + prove that Model ⊨ Spec.

The three ingredients are inseparable: a model without a spec is meaningless, a spec without verification is wishful thinking.

# The Verification Spectrum

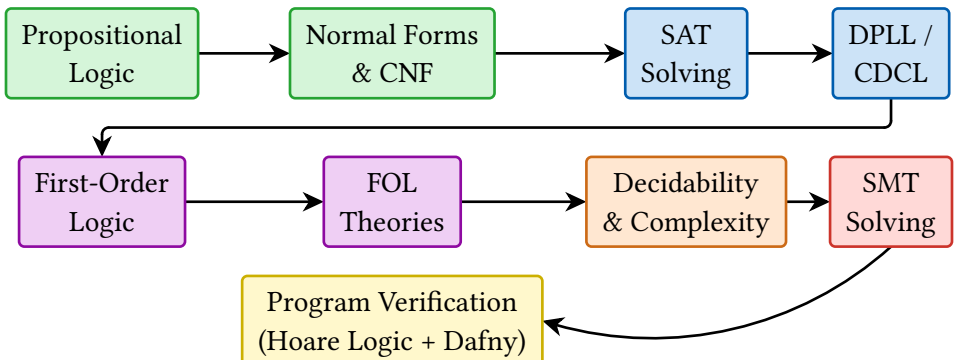
| Method | Rigor | Coverage | Cost | Proves $M \models S$? |
|---|---|---|---|---|
| Testing | Low | Partial | Low | No |
| Static analysis | Medium | Heuristic | Low | Partially |
| Model checking | High | Exhaustive (bounded) | Medium | Yes (bounded) |
| Deductive verification | Highest | Complete | High | Yes |

This course moves from left to right, ending with deductive verification in Dafny.

> *"Testing shows the presence, not the absence of bugs."* — Edsger W. Dijkstra (1969)

# Course Roadmap

*"How do we make machines reason about correctness?"*



# Propositional Logic Refresher

# Syntax vs Semantics: Why Two Perspectives?

Logic has two faces: *syntax* (the strings we write and transform) and *semantics* (what they *mean*).

**Syntactic World ($\vdash$)**

- Formulas, rewriting rules
- Proof systems, derivations
- *"I can derive $\alpha$ from $\Gamma$"*
- Symbol: $\Gamma \vdash \alpha$

**Semantic World ($\vDash$)**

- Interpretations, truth values
- Truth tables, models
- *"$\alpha$ is true whenever $\Gamma$ is"*
- Symbol: $\Gamma \vDash \alpha$

For propositional logic, these two worlds are *perfectly aligned* — soundness + completeness gives us $\vdash \Longleftrightarrow \vDash$. So why bother distinguishing them?

**Reason 1 — Different algorithms:** Semantics gives *truth tables* ($2^n$ rows — brute force). Syntax gives *proof search* (sometimes exponentially shorter). For 300 variables, a truth table has $2^{300}$ rows (more than atoms in the universe), but a proof might take 50 lines. Same question, vastly different computational cost.

**Reason 2 — The gap appears later:** For first-order logic over arithmetic, Gödel's Incompleteness Theorem shows there are true statements that *no* proof system can derive: $\vdash \subsetneq \vDash$. When this gap opens, confusing the two perspectives leads to fundamental errors.

# Syntax vs Semantics: Why Two Perspectives? [2]

**Bottom line:**

- Semantics asks *"Is it true?"* (check all interpretations).
- Syntax asks *"Can I derive it?"* (apply inference rules mechanically).

For PL, both always give the same answer.

However, we *train the distinction now* so it is natural when it matters.

# PL Syntax

Propositional logic studies *Boolean combinations* of atomic statements.
Its syntax defines which strings are "legal" formulas:

**Definition 4** (Well-Formed Formula (WFF)): Given propositional variables $P, Q, R, \ldots$ and constants $\top, \bot$, the set of *well-formed formulas* is defined inductively:

1. Every propositional variable and constant is a WFF.
2. If $\alpha$ and $\beta$ are WFFs, then $\neg\alpha$, $(\alpha \wedge \beta)$, $(\alpha \vee \beta)$, $(\alpha \rightarrow \beta)$, $(\alpha \iff \beta)$ are WFFs.
3. Nothing else is a WFF.

**Conventions:**
Operator precedence: $\neg > \wedge > \vee > \rightarrow > \iff$.
Outer parentheses omitted. Associativity: $\wedge$, $\vee$ left-to-right; $\rightarrow$ right-to-left.

*Example*: A Boolean guard `if (x > 0 && !done)` in a program corresponds to the propositional formula $P \wedge \neg Q$, where $P$ stands for `x > 0` and $Q$ for `done`. This is a WFF by rule 2.

# PL Semantics

**Definition 5**: An *interpretation* (valuation) $\nu : V \to \{0, 1\}$ assigns a truth value to each propositional variable.

The *evaluation* $[\![\alpha]\!]_\nu$ of a formula $\alpha$ under $\nu$ is defined recursively:

$$
\begin{aligned}
[\![\top]\!]_\nu = 1, \quad [\![\bot]\!]_\nu = 0, \quad [\![P]\!]_\nu &= \nu(P) \\
[\![\neg\alpha]\!]_\nu &= 1 - [\![\alpha]\!]_\nu \\
[\![\alpha \wedge \beta]\!]_\nu &= \min([\![\alpha]\!]_\nu, [\![\beta]\!]_\nu) \\
[\![\alpha \vee \beta]\!]_\nu &= \max([\![\alpha]\!]_\nu, [\![\beta]\!]_\nu) \\
[\![\alpha \to \beta]\!]_\nu &= \max(1 - [\![\alpha]\!]_\nu, [\![\beta]\!]_\nu)
\end{aligned}
$$

**Definition 6**: An interpretation $\nu$ *satisfies* a formula $\alpha$, written $\nu \vDash \alpha$, if $[\![\alpha]\!]_\nu = 1$.

A *model* of $\alpha$ is any interpretation that satisfies it.

# PL Semantics [2]

**Terminology note:** The word "model" appears in many distinct contexts:

- **PL model** = an interpretation (truth assignment) satisfying a formula.
- **FOL model** = a structure (domain + interpretation of symbols) satisfying sentences.
- **Model checking** = algorithmic verification technique (checking if a system model satisfies a temporal property).

In this course, context determines which meaning applies.
For now, "model" = satisfying interpretation.

*Example*: Let $\nu(P) = 1, \nu(Q) = 0$.

- Then $[\![P \to Q]\!]_\nu = \max(1-1, 0) = 0$ and $[\![\neg P \vee Q]\!]_\nu = \max(0, 0) = 0$.
- Both agree, as expected from the equivalence $(P \to Q) \equiv (\neg P \vee Q)$.

Since $[\![P \to Q]\!]_\nu = 0$, we say $\nu \not\models (P \to Q)$ — this interpretation is *not* a model of $P \to Q$.

# PL Semantics [3]

*Example*: Let $\nu(A) = 1, \nu(B) = 0, \nu(C) = 1$. Evaluate $A \wedge (B \vee C)$:

$$[\![B \vee C]\!]_\nu = \max(0, 1) = 1$$
$$[\![A \wedge (B \vee C)]\!]_\nu = \min(1, 1) = 1$$

The formula is *satisfied* by this interpretation: $\nu \vDash A \wedge (B \vee C)$. So $\nu$ is a *model* of $A \wedge (B \vee C)$.

# Semantic Classification

Formulas are classified by their truth behavior across *all* interpretations:

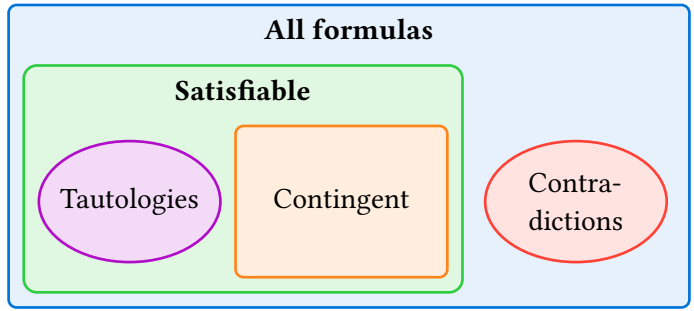
**Definition 7** (Semantic Classification): Let $\alpha$ be a WFF.

- $\alpha$ is **valid** (*tautology*), written $\models \alpha$, if *every* interpretation is a model: $\nu \models \alpha$ for all $\nu$.
- $\alpha$ is **satisfiable** (*consistent*) if it has *at least one* model: $\nu \models \alpha$ for some $\nu$.
- $\alpha$ is **unsatisfiable** (*contradiction*) if it has *no* models: $\nu \not\models \alpha$ for all $\nu$.
- $\alpha$ is **falsifiable** if some interpretation is *not* a model: $\nu \not\models \alpha$ for some $\nu$.

*Example*:

- $P \vee \neg P$ — valid (tautology). *Every* interpretation is a model.
- $P \wedge Q$ — satisfiable (has model $\nu(P) = \nu(Q) = 1$) and falsifiable (non-model $\nu(P) = 1, \nu(Q) = 0$). This is *contingent*.
- $P \wedge \neg P$ — unsatisfiable. *No* model exists.

# Semantic Classification [2]



# Entailment vs Implication

**Definition 8** (Semantic Entailment): A set of formulas $\Gamma$ *semantically entails* $\alpha$, written $\Gamma \vDash \alpha$, if every model of $\Gamma$ is also a model of $\alpha$.

Equivalently: every interpretation satisfying all formulas in $\Gamma$ also satisfies $\alpha$.

These two notions — *implication* and *entailment* — are distinct but deeply related:

**Implication (→)** is a *connective* — an operator *inside* the language of propositional logic.

- $P \to Q$ is a well-formed formula with a truth value under each interpretation.
- It can appear in compound formulas: $(P \to Q) \land R$, $\neg(P \to Q)$, *etc.*
- Defined by a truth table: $[\![P \to Q]\!]_\nu = \max(1 - [\![P]\!]_\nu, [\![Q]\!]_\nu)$.

**Entailment (⊨)** is a *metalogical* relation — a claim *about* formulas from outside the logic.

- $P \vDash Q$ is *not* a formula; it is a mathematical statement about all interpretations.
- It cannot be negated or combined using logical connectives.
- Defined by quantifying over models: *"every model of $P$ is also a model of $Q$."*

# Entailment vs Implication [2]

**Common mistake:** writing $P \vDash Q$ when you mean $P \to Q$, or vice versa.

One ($\to$) is a formula you can evaluate; the other ($\vDash$) is a claim you prove.

**Why distinguish them?** In propositional logic, they coincide (via the Deduction Theorem). But in first-order logic and beyond, the distinction becomes crucial: some truths are not provable, and the syntactic world ($\vdash$) diverges from the semantic world ($\vDash$).

For PL, these two worlds coincide perfectly: " $\vdash$ " $\iff$ " $\vDash$ ". For first-order logic, Gödel showed that in specific theories (like Peano Arithmetic), *true* sentences can be *unprovable* — the syntactic and semantic worlds diverge. We will see this precisely in the Metatheorems section.

# Entailment vs Implication [3]

**Theorem 1** (Deduction Theorem (Semantic)): For any formulas $\alpha, \beta$:

$$\alpha \vDash \beta \quad \Longleftrightarrow \quad \vDash \alpha \to \beta$$

**Proof** $(\Rightarrow)$: Assume $\alpha \vDash \beta$. We must show $\vDash \alpha \to \beta$.

Let $\nu$ be any interpretation. We show $\nu \vDash \alpha \to \beta$.

- If $\nu \nvDash \alpha$, then $[\![\alpha \to \beta]\!]_\nu = \max(0, [\![\beta]\!]_\nu) = 1$, so $\nu \vDash \alpha \to \beta$.
- If $\nu \vDash \alpha$, then since $\alpha \vDash \beta$, we have $\nu \vDash \beta$, so $[\![\alpha \to \beta]\!]_\nu = \max(0, 1) = 1$.

In both cases, $\nu \vDash \alpha \to \beta$. Since $\nu$ was arbitrary, $\alpha \to \beta$ is valid. $\square$

**Proof** $(\Leftarrow)$: Assume $\vDash \alpha \to \beta$. We must show $\alpha \vDash \beta$.

Let $\nu$ be a model of $\alpha$, *i.e.*, $\nu \vDash \alpha$. Since $\alpha \to \beta$ is valid, $\nu \vDash \alpha \to \beta$. By the definition of $\to$: $\max(1 - [\![\alpha]\!]_\nu, [\![\beta]\!]_\nu) = 1$. Since $[\![\alpha]\!]_\nu = 1$, we have $\max(0, [\![\beta]\!]_\nu) = 1$, so $[\![\beta]\!]_\nu = 1$, *i.e.*, $\nu \vDash \beta$.

Thus every model of $\alpha$ is a model of $\beta$. $\square$

# Entailment vs Implication [4]

The Deduction Theorem connects semantic entailment to formula validity — it lets us reduce the question *"does $\alpha$ entail $\beta$?"* to *"is $\alpha \to \beta$ valid?"* This reduction is what makes automated validity checking possible: entailment becomes a satisfiability check.

*Example*: $\{P, P \to Q\} \vDash Q$ (modus ponens as entailment)

**Via Deduction Theorem:** This is equivalent to $P \land (P \to Q) \to Q$ being valid.

Check: any interpretation either falsifies $P \land (P \to Q)$ (making the implication vacuously true), or satisfies both $P$ and $P \to Q$, in which case it must satisfy $Q$ (by the truth table for $\to$).

*Example*: $\neg(P \land Q) \vDash \neg P \lor \neg Q$ (De Morgan, semantic form)

**Via Deduction Theorem:** Equivalent to $\vDash \neg(P \land Q) \to (\neg P \lor \neg Q)$, which is a tautology.

# Entailment vs Implication [5]

**Theorem 2** (Generalized Deduction Theorem): For any set of formulas $\Gamma$ and formulas $\alpha, \beta$:

$$\Gamma \cup \{\alpha\} \vDash \beta \quad \Longleftrightarrow \quad \Gamma \vDash \alpha \to \beta$$

**Proof** *($\Rightarrow$)*: Assume $\Gamma \cup \{\alpha\} \vDash \beta$. Let $\nu$ be a model of $\Gamma$. We show $\nu \vDash \alpha \to \beta$.

- If $\nu \nvDash \alpha$, then $\nu \vDash \alpha \to \beta$ (vacuously).
- If $\nu \vDash \alpha$, then $\nu$ models all formulas in $\Gamma \cup \{\alpha\}$, so by assumption $\nu \vDash \beta$, hence $\nu \vDash \alpha \to \beta$. □

**Proof** *($\Leftarrow$)*: Assume $\Gamma \vDash \alpha \to \beta$. Let $\nu$ be a model of $\Gamma \cup \{\alpha\}$. Then $\nu \vDash \Gamma$, so $\nu \vDash \alpha \to \beta$ by assumption. Since $\nu \vDash \alpha$, we have $\nu \vDash \beta$ by modus ponens. □

**Note**: This theorem justifies the *hypothetical reasoning* pattern: to show $\Gamma \vDash \alpha \to \beta$, it suffices to show $\Gamma \cup \{\alpha\} \vDash \beta$ — *i.e.*, *"assume $\alpha$ as an additional hypothesis and derive $\beta$."*

# SAT vs VALID Duality

Satisfiability and validity are *dual* decision problems:

$$\begin{aligned} \text{SAT:} \quad & \exists \nu.\, \nu \vDash \alpha \quad \text{(find a model)} \\ \text{VALID:} \quad & \forall \nu.\, \nu \vDash \alpha \quad \text{(every interpretation is a model)} \end{aligned}$$

$$\alpha \text{ is valid} \quad \Longleftrightarrow \quad \neg\alpha \text{ is unsatisfiable.}$$

*Example*: $P \lor \neg P$ is valid $\Longleftrightarrow \neg(P \lor \neg P) \equiv P \land \neg P$ is unsatisfiable.

Checking SAT by truth tables takes $\mathcal{O}(2^n)$ time.
Is there a polynomial algorithm? *This is the P vs NP problem* — a Millennium Prize question.

# Fundamental Equivalence Laws

$\alpha \equiv \beta$ iff $\alpha \iff \beta$ is a tautology.

These equivalences form the *toolkit* for normal form transformations. Every conversion (NNF, CNF, DNF) is a sequence of applications of these rewriting rules:

**Double Negation:**

- $\neg\neg A \equiv A$

**De Morgan's Laws:**

- $\neg(A \wedge B) \equiv \neg A \vee \neg B$
- $\neg(A \vee B) \equiv \neg A \wedge \neg B$

**Implication:**

- $(A \to B) \equiv (\neg A \vee B)$

**Distributivity:**

- $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$
- $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$

**Contraposition:**

- $(A \to B) \equiv (\neg B \to \neg A)$

**Identity:**

- $A \wedge \top \equiv A$
- $A \vee \bot \equiv A$

**Complement:**

- $A \wedge \neg A \equiv \bot$
- $A \vee \neg A \equiv \top$

**Exportation:**

- $(A \wedge B) \to C \equiv A \to (B \to C)$

# Completeness of Connective Sets

For $n$ Boolean variables, there are $2^{2^n}$ possible Boolean functions.

How many connectives do we *really* need?

**Definition 9** (Functional Completeness): A set $S$ of connectives is *functionally complete* if every Boolean function can be expressed using only connectives from $S$.

*Example*:

- $\{\neg, \wedge, \vee\}$ — the standard Boolean basis.
- $\{\neg, \wedge\}$ — And-Inverter Graphs (AIGs), used in hardware verification.
- $\{\neg, \vee\}$
- $\{\overline{\wedge}\}$ — NAND alone suffices.
- $\{\overline{\vee}\}$ — NOR alone suffices.

*Example ($\{\wedge, \rightarrow\}$ is not complete.)*: Let $\alpha$ be any WFF using only $\wedge$ and $\rightarrow$, and let $\nu$ assign 1 to every variable. By structural induction, $[\![\alpha]\!]_\nu = 1$ for all such $\alpha$.
But $\neg P$ evaluates to 0 under this $\nu$ — so $\neg P$ is not expressible.

# Completeness of Connective Sets [2]

**Why FM cares:** In hardware verification, circuits are built from NAND/NOR gates — completeness guarantees these gates can implement *any* Boolean function. In SAT solving, CNF uses only $\{\neg, \wedge, \vee\}$ — so no expressiveness is lost.

# Normal Forms

# Why Normal Forms?

We know PL formulas can express any Boolean function. But SAT solvers don't accept *arbitrary* formulas — they need a standardized input format. Normal forms provide exactly this: every formula is rewritten into a restricted shape that algorithms can uniformly process.

Three normal forms, each with different trade-offs:

- **Negation Normal Form (NNF)** — negations pushed to atoms; cheap to compute, preserves structure
- **Conjunctive Normal Form (CNF)** — conjunction of clauses; *the language of SAT solvers*
- **Disjunctive Normal Form (DNF)** — disjunction of cubes; dual of CNF

Every propositional formula can be converted to an *equivalent* formula in any of these forms. The key question is: *at what cost?*

# Literals and Their Complements

**Definition 10** (Literal): A *literal* is a propositional variable ($p$ — *positive*) or its negation ($\neg p$ — *negative*).

**Definition 11** (Complement): The *complement* of a literal $\ell$ is denoted $\overline{\ell}$:

$$\overline{\ell} = \begin{cases} \neg p & \text{if } \ell \equiv p \quad \text{(positive)} \\ p & \text{if } \ell \equiv \neg p \quad \text{(negative)} \end{cases}$$

Complementary literals $\ell$ and $\overline{\ell}$ always satisfy $\ell \wedge \overline{\ell} \equiv \bot$ and $\ell \vee \overline{\ell} \equiv \top$.

# Clauses and Cubes

**Definition 12** (Clause): A *clause* is a disjunction of literals: $\ell_1 \lor \ell_2 \lor \ldots \lor \ell_k$.

- An *empty clause* $\square$ contains no literals and is *unsatisfiable* (false in every interpretation).
- A *unit clause* contains exactly one literal.
- A *Horn clause* contains at most one positive literal.

**Definition 13** (Cube): A *cube* is a conjunction of literals: $\ell_1 \land \ell_2 \land \ldots \land \ell_k$.

Clauses and cubes are *dual* notions:

- A clause is *falsified* only if *every* literal in it is false.
- A cube is *satisfied* only if *every* literal in it is true.

**Horn clauses** are computationally special: SAT restricted to Horn clauses is solvable in *linear time* via unit propagation. Prolog's inference engine works exclusively with Horn clauses.

# Negation Normal Form

**Definition 14** (Negation Normal Form (NNF)): A formula is in NNF if:

1. Negation ($\neg$) is applied only to *atoms* (propositional variables).
2. The only connectives are $\wedge$, $\vee$, and $\neg$ (applied to atoms).

**Grammar:**

$$\begin{aligned}
\langle \text{Atom} \rangle &::= \top \mid \bot \mid \langle \text{Variable} \rangle \\
\langle \text{Literal} \rangle &::= \langle \text{Atom} \rangle \mid \neg \langle \text{Atom} \rangle \\
\langle \text{Formula} \rangle &::= \langle \text{Literal} \rangle \mid \langle \text{Formula} \rangle \wedge \langle \text{Formula} \rangle \mid \langle \text{Formula} \rangle \vee \langle \text{Formula} \rangle
\end{aligned}$$

In words: only $\wedge$, $\vee$, and negation applied directly to variables — no $\rightarrow$, no $\Longleftrightarrow$, no nested $\neg$.

*Example*:

- $(p \wedge q) \vee (\neg p \wedge \neg q)$ — in NNF.
- $\neg(p \wedge q)$ — *not* in NNF (negation applied to a compound formula).

# NNF Transformation

Rewriting rules (apply until no rule matches):

| **Description** | **Rewrite rule** |
|---|---|
| Eliminate implications | $(A \to B) \Longrightarrow (\neg A \lor B)$ |
| Eliminate biconditionals | $(A \Longleftrightarrow B) \Longrightarrow (\neg A \lor B) \land (A \lor \neg B)$ |
| De Morgan (conjunction) | $\neg(A \land B) \Longrightarrow (\neg A \lor \neg B)$ |
| De Morgan (disjunction) | $\neg(A \lor B) \Longrightarrow (\neg A \land \neg B)$ |
| Double negation | $\neg\neg A \Longrightarrow A$ |

# NNF Transformation [2]

**Theorem 3**: Every formula *not containing* $\iff$ can be converted to an equivalent NNF with a *linear increase* in size.

Formulas *containing* $\iff$ may suffer *exponential blowup* when converted to NNF.

Why the blowup? The biconditional $A \iff B$ expands to $(\neg A \lor B) \land (A \lor \neg B)$ — producing *two copies* of $A$ and $B$. A chain of $n$ biconditionals $p_1 \iff p_2 \iff \ldots \iff p_n$ doubles the formula at each level, producing $2^n$ copies. The Tseitin transformation (coming soon) avoids this.

# NNF Transformation: Worked Example

Convert $(P \to Q) \to R$ to NNF step by step:

$$
\begin{aligned}
& (P \to Q) \to R \\
\Longrightarrow\ & \neg(P \to Q) \vee R && \text{(eliminate outer } \to \text{)} \\
\Longrightarrow\ & \neg(\neg P \vee Q) \vee R && \text{(eliminate inner } \to \text{)} \\
\Longrightarrow\ & (\neg\neg P \wedge \neg Q) \vee R && \text{(De Morgan)} \\
\Longrightarrow\ & (P \wedge \neg Q) \vee R && \text{(Double negation)}
\end{aligned}
$$

Result: $(P \wedge \neg Q) \vee R$ — negations only on atoms.

# Conjunctive Normal Form

**Definition 15** (Conjunctive Normal Form (CNF)): A formula is in CNF if it is a conjunction of clauses:

$$\alpha = \bigwedge_i \bigvee_j \ell_{ij}$$

*Example*: $\alpha = (\neg p \vee q) \wedge (\neg p \vee q \vee r) \wedge (\neg q)$ — CNF with 3 clauses.

**Why CNF?** Every modern SAT solver (MiniSat, CaDiCaL, Kissat) operates on CNF. Satisfaction requires *at least one* literal per clause — this "one per clause" structure is what makes unit propagation and resolution work.

# Disjunctive Normal Form

**Definition 16** (Disjunctive Normal Form (DNF)): A formula is in DNF if it is a disjunction of cubes:

$$\alpha = \bigvee_i \bigwedge_j \ell_{ij}$$

*Example*: $\alpha = (p \wedge q) \vee (\neg p \wedge q \wedge r) \vee (\neg q)$ — DNF with 3 cubes.

**CNF vs DNF — dual complexities:**

| Problem | On CNF | On DNF |
|---|---|---|
| SAT check | NP-complete | Polynomial |
| VALID check | Polynomial | co-NP-complete |

- SAT on DNF is polynomial: check if any cube has no complementary literals.
- VALID on CNF is polynomial: check if every clause contains complementary literals.

# CNF Transformation

Any formula can be converted to CNF:

1. Apply NNF transformation rules.
2. Distribute $\vee$ over $\wedge$ (flattening):
   - $A \vee (B \wedge C) \Longrightarrow (A \vee B) \wedge (A \vee C)$
   - $(A \wedge B) \vee C \Longrightarrow (A \vee C) \wedge (B \vee C)$
3. Normalize: flatten nested $\wedge$ and $\vee$.

**Theorem 4**: Every formula can be converted to an *equivalent* CNF, but the size may grow *exponentially*.

The *distributive law* is the culprit:

$$\underbrace{(x_1 \wedge y_1) \vee (x_2 \wedge y_2) \vee \ldots \vee (x_n \wedge y_n)}_{n \text{ cubes}} \overset{\text{CNF}}{\Longrightarrow} \underbrace{\ldots}_{2^n \text{ clauses}}$$

**Question:** Can we avoid this blowup? *Yes — by relaxing "equivalence" to "equisatisfiability".*

# Equisatisfiability

**Definition 17**: Two formulas $\alpha$ and $\beta$ are *equisatisfiable* if $\alpha$ is satisfiable iff $\beta$ is satisfiable.

**Note**: Equisatisfiability is *weaker* than logical equivalence. For SAT solving, equisatisfiability suffices — we only care *whether* a satisfying assignment exists. Any model of the equisatisfiable formula can be *restricted* to the original variables.

*Example*: $P \wedge Q$ and $(P \wedge Q) \wedge (n \iff P)$ are equisatisfiable but *not* equivalent — the second formula has an extra variable $n$ and is defined over a strictly larger language.

# Tseitin Transformation

**Definition 18**: The *Tseitin transformation* converts any formula to CNF in *polynomial time* by introducing *fresh* variables.

For each non-literal subformula $A$ of a formula $F$:

1. Introduce a fresh propositional variable $n_A$.
2. Add a *definitional clause* $n_A \iff A$ (asserting equivalence).
3. Replace $A$ with $n_A$ in $F$.

The resulting formula is *equisatisfiable* with the original:

- Every model of $F$ extends to a model of the Tseitin encoding.
- Every model of the encoding restricted to original variables satisfies $F$.

# Tseitin Transformation [2]

## □ Cost

- $\mathcal{O}(n)$ fresh variables and $\mathcal{O}(n)$ clauses, where $n$ is the formula size.
- Each definitional clause $n \iff A$ for a binary connective produces a *constant number* of clauses.

*Example*: The definition $n \iff (A \wedge B)$ is equivalent to:

$$(n \to (A \wedge B)) \wedge ((A \wedge B) \to n) \quad \equiv \quad (\neg n \vee A) \wedge (\neg n \vee B) \wedge (\neg A \vee \neg B \vee n)$$

That is: 3 clauses.

# Tseitin Transformation: Example

*Example*: $F = p_1 \iff (p_2 \iff (p_3 \iff (p_4 \iff (p_5 \iff p_6))))$

**Equivalent CNF**: $2^5 = 32$ clauses (exponential).

**Tseitin transformation**: introduce fresh variables $n_3, n_4, n_5$:

$$\begin{aligned} S = p_1 &\iff (p_2 \iff n_3) \land \\ n_3 &\iff (p_3 \iff n_4) \land \\ n_4 &\iff (p_4 \iff n_5) \land \\ n_5 &\iff (p_5 \iff p_6) \end{aligned}$$

Each biconditional definition produces a constant number of clauses (4 clauses for $\iff$).

**Equisatisfiable CNF**: 16 clauses, 3 fresh variables. *Linear growth.*

# Clausal Form

**Definition 19**: A *clausal form* of a formula $F$ is a *set* of clauses $S_F$ which is satisfiable iff $F$ is satisfiable. Moreover, $F$ and $S_F$ have the same models when restricted to the language of $F$.

The main advantage: any formula can be converted to clausal form in *almost linear* time.

**Algorithm:**

1. If $F = C_1 \wedge \ldots \wedge C_n$ where each $C_i$ is already a clause, then $S_F = \{C_1, \ldots, C_n\}$.
2. Otherwise, apply Tseitin transformation: name each non-literal subformula with a fresh variable.

**Key insight:** The clausal form is the bridge between arbitrary formulas and SAT solvers. It preserves satisfiability while keeping the representation compact.

# Proof Systems

# Why Proof Systems?

Clausal form gives us the *input format* for SAT solvers. But before we build solvers (next lecture), we need to understand what it means to *prove* things mechanically — that is the purpose of proof systems.

Truth tables require $2^n$ rows for $n$ variables. For 300 variables (modest by industrial standards), that exceeds the number of atoms in the universe.

Proof systems *derive* validity step by step using inference rules. A clever proof can be *exponentially shorter* than brute-force enumeration.

**Definition 20**: A **proof system** derives valid formulas (or entailments) by applying *inference rules* to *axioms* and *assumptions*.

Three main traditions:

- **Hilbert-style:** many axiom schemas, one rule (modus ponens). Compact to define, hard to use.
- **Natural deduction** (Gentzen, 1934): no axioms, intro/elim rules per connective. *Our primary tool.*
- **Sequent calculus** (Gentzen, 1934): manipulates structured judgments. Foundation of automated proof search.

# Natural Deduction

A proof system with *no axioms* — only inference rules. Each connective has *introduction* rules (**how to build** a compound formula) and *elimination* rules (**how to use** one).

We present proofs in **Fitch notation**: a numbered list of steps. Each step contains a formula and a *justification* (the rule applied + referenced line numbers).

*Subproofs* (indented blocks) introduce a *temporary assumption*. Everything derived inside a subproof depends on that assumption. When the subproof closes, the assumption is *discharged* — you may no longer cite its internal lines, but you can reference the subproof *as a whole*.

**Mental model:** A subproof says *"if I temporarily assume $\alpha$, I can derive $\beta$."* When it closes, you conclude $\alpha \to \beta$ — without assuming $\alpha$ anymore.

# Conjunction and Implication Rules

**$\wedge$-introduction** ($\wedge$i):
From $\alpha$ and $\beta$ on separate lines, conclude $\alpha \wedge \beta$.

| | | |
|---|---|---|
| 1 | $\alpha$ | *Premise* |
| 2 | $\beta$ | *Premise* |
| 3 | $\alpha \wedge \beta$ | $\wedge i$ *1, 2* |

**$\wedge$-elimination** ($\wedge$e):
From $\alpha \wedge \beta$, conclude $\alpha$ (or $\beta$).

| | | |
|---|---|---|
| 1 | $\alpha \wedge \beta$ | *Premise* |
| 2 | $\alpha$ | $\wedge e$ *1* |
| 3 | $\beta$ | $\wedge e$ *1* |

# Conjunction and Implication Rules [2]

**$\rightarrow$-introduction** ($\rightarrow$i):
Open a subproof assuming $\alpha$, derive $\beta$.
Close subproof, conclude $\alpha \rightarrow \beta$.

| | | |
|---|---|---|
| 1 | $\alpha$ | *assumption* |
| 2 | $\vdots$ | |
| 3 | $\beta$ | $\vdots$ |
| 4 | $\alpha \rightarrow \beta$ | $\rightarrow$*i 1–3* |

**$\rightarrow$-elimination** ($\rightarrow$e):
Modus ponens.
From $\alpha$ and $\alpha \rightarrow \beta$, conclude $\beta$.

| | | |
|---|---|---|
| 1 | $\alpha$ | *Premise* |
| 2 | $\alpha \rightarrow \beta$ | *Premise* |
| 3 | $\beta$ | $\rightarrow$*e 1, 2* |

# Disjunction Rules

**$\vee$-introduction** ($\vee$i):
From $\alpha$, conclude $\alpha \vee \beta$ (or $\beta \vee \alpha$).

| | | |
|---|---|---|
| 1 | $\alpha$ | *Premise* |
| 2 | $\alpha \vee \beta$ | $\vee i$ *1* |

**$\vee$-elimination** ($\vee$e):
From $\alpha \vee \beta$, with subproofs deriving $\gamma$ from each disjunct, conclude $\gamma$.

| | | |
|---|---|---|
| 1 | $\alpha \vee \beta$ | *Premise* |
| 2 | $\quad \alpha$ | *assumption* |
| 3 | $\quad \gamma$ | $\vdots$ |
| 4 | $\quad \beta$ | *assumption* |
| 5 | $\quad \gamma$ | $\vdots$ |
| 6 | $\gamma$ | $\vee e$ *1, 2–3, 4–5* |

# Negation and Absurdity Rules

**$\neg$-introduction** ($\neg$i):
Assume $\alpha$, derive $\bot$ (contradiction).
Close subproof, conclude $\neg\alpha$.

| | | |
|---|---|---|
| 1 | $\alpha$ | *assumption* |
| 2 | $\vdots$ | |
| 3 | $\bot$ | $\vdots$ |
| 4 | $\neg\alpha$ | $\neg i$ *1–3* |

**$\neg$-elimination** ($\neg$e):
From $\alpha$ and $\neg\alpha$, derive $\bot$.

| | | |
|---|---|---|
| 1 | $\alpha$ | *Premise* |
| 2 | $\neg\alpha$ | *Premise* |
| 3 | $\bot$ | $\neg e$ *1, 2* |

# Negation and Absurdity Rules [2]

**$\bot$-elimination** (ex falso quodlibet, $\bot$e):
From $\bot$, derive *any* formula.

| | | |
|---|---|---|
| 1 | $\bot$ | *Premise* |
| 2 | $\alpha$ | *$\bot$e 1* |

**Reductio ad absurdum** (RAA):
Assume $\neg\alpha$, derive $\bot$. Conclude $\alpha$.

| | | |
|---|---|---|
| 1 | $\neg\alpha$ | *assumption* |
| 2 | $\vdots$ | |
| 3 | $\bot$ | $\vdots$ |
| 4 | $\alpha$ | *RAA 1–3* |

**Classical vs Intuitionistic:** RAA and LEM ($\alpha \vee \neg\alpha$) are *classical* rules. Dropping them gives *intuitionistic* logic, where $P \vee \neg P$ is not provable.

**Two key proof patterns** that cover the majority of ND proofs:

- To prove $\neg\varphi$: assume $\varphi$, derive $\bot$, apply $\neg$i.
- To prove $\alpha \to \beta$: assume $\alpha$, derive $\beta$, apply $\to$i.

# Fitch Proofs: Basic Examples

*Example*: **Conjunction rearrangement:** $p \wedge q, r \vdash q \wedge r$

| | | |
|---|---|---|
| 1 | $p \wedge q$ | *Premise* |
| 2 | $r$ | *Premise* |
| 3 | $q$ | $\wedge e$ *1* |
| 4 | $q \wedge r$ | $\wedge i$ *3, 2* |

Each step cites the rule and the line numbers it depends on. Line 3 *eliminates* the conjunction to extract $q$; line 4 *introduces* a new conjunction.

# Fitch Proofs: Basic Examples [2]

*Example*: **Modus Tollens:** $A \to B, \neg B \vdash \neg A$

| | | |
|---|---|---|
| 1 | $A \to B$ | *Premise* |
| 2 | $\neg B$ | *Premise* |
| 3 | $\quad A$ | *assumption* |
| 4 | $\quad B$ | $\to$*e 3, 1* |
| 5 | $\quad \bot$ | $\neg$*e 4, 2* |
| 6 | $\neg A$ | $\neg$*i 3–5* |

Lines 3–5 form a *subproof*: we temporarily assume $A$, derive $\bot$, then discharge the assumption to conclude $\neg A$ via $\neg$i. The vertical bar shows the scope of the assumption — lines 4 and 5 are only accessible *within* the subproof.

# Fitch Proofs: Implication Chains

*Example*: **Hypothetical Syllogism:** $A \to B, B \to C \vdash A \to C$

| | | |
|---|---|---|
| 1 | $A \to B$ | *Premise* |
| 2 | $B \to C$ | *Premise* |
| 3 | $\quad A$ | *assumption* |
| 4 | $\quad B$ | $\to$*e 3, 1* |
| 5 | $\quad C$ | $\to$*e 4, 2* |
| 6 | $A \to C$ | $\to$*i 3–5* |

To prove an implication $A \to C$, we *assume* $A$ (line 3), derive $C$ (line 5), and close with $\to$i.

**Note**: More worked Fitch proofs (disjunctive syllogism, De Morgan) appear in the exercises. The key patterns: $\vee$-elimination requires two subproofs (one per disjunct); $\neg$-introduction assumes $\varphi$, derives $\bot$, concludes $\neg\varphi$.

# Soundness and Completeness

**Definition 21** (Soundness): A proof system is *sound* if every provable formula is valid:

$$\text{If } \Gamma \vdash \alpha \text{ then } \Gamma \vDash \alpha. \qquad \text{(Nothing false is provable.)}$$

**Definition 22** (Completeness): A proof system is *complete* if every valid formula is provable:

$$\text{If } \Gamma \vDash \alpha \text{ then } \Gamma \vdash \alpha. \qquad \text{(Nothing true is unprovable.)}$$

**Theorem 5**: Propositional natural deduction is both *sound* and *complete*.

$$\Gamma \vdash \alpha \iff \Gamma \vDash \alpha$$

**Proof**: *(Soundness.)* By induction on the derivation. Each inference rule preserves validity: a small truth-table check per rule.

# Soundness and Completeness [2]

*(Completeness.)* Build a derivation by induction on $\alpha$'s structure, case-splitting on which variables $\Gamma$ forces.
*(Kalmár, 1935.)* $\square$

Soundness: verified properties *actually hold*. Completeness: every true property *can* be proven.

# Proof Strategies

**Forward reasoning** (bottom-up):

- Start from premises $\Gamma$
- Apply rules to derive new facts
- Continue until $\alpha$ is derived
- Risk: combinatorial explosion

**Refutation** (top-down):

- Assume $\neg\alpha$ together with $\Gamma$
- Derive a contradiction ($\bot$)
- Conclude that $\alpha$ must hold
- Advantage: *goal-directed* search

Refutation is the basis for *automated* reasoning: searching for a contradiction in $\Gamma \cup \{\neg\alpha\}$ is equivalent to a SAT problem.

**Other proof systems**: *Semantic tableaux* (truth trees) are another refutation-based method: negate the goal, decompose formulas, and check if every branch closes. Open branches yield counterexamples. We skip the details — resolution (below) is more directly relevant to SAT solving.

**Bridge to automated reasoning:** Resolution is the foundation of SAT solvers. Unlike natural deduction (designed for humans), resolution has a single rule — perfect for implementation.

# Resolution

*Resolution* reduces propositional proof theory to a single inference rule, but requires *clausal form* (CNF). This makes it the natural foundation for automated theorem proving.

**Definition 23** (Resolution Rule): Given two clauses containing complementary literals:

$$C_1 = (\ell_1 \vee \ldots \vee \ell_m \vee p) \quad \text{and} \quad C_2 = (\ell'_1 \vee \ldots \vee \ell'_k \vee \neg p)$$

derive the *resolvent*:

$$C = C_1 \otimes_p C_2 = \ell_1 \vee \ldots \vee \ell_m \vee \ell'_1 \vee \ldots \vee \ell'_k$$

The variable $p$ is called the *pivot*.

*Example*: $(\neg P \vee Q)$ and $(P \vee R)$ resolve on $P$ to produce $(Q \vee R)$.

Resolution is a *refutation* system: to prove $\Gamma \vDash \alpha$, convert $\Gamma \cup \{\neg\alpha\}$ to CNF and derive the *empty clause* $\square$.

# Resolution Refutation: Example

**Prove by resolution:** $\{P, P \to Q\} \vDash Q$.

Add $\neg Q$ (negation of goal) and convert to clauses:

$$\begin{aligned} C_1 &= \{P\} && \text{(from } P \text{)} \\ C_2 &= \{\neg P, Q\} && \text{(from } P \to Q \equiv \neg P \vee Q \text{)} \\ C_3 &= \{\neg Q\} && \text{(negation of goal)} \end{aligned}$$

Derive:

$$\begin{aligned} C_4 &= \{Q\} \text{ resolve } C_1 \text{ and } C_2 \text{ on } P \\ C_5 &= \square \quad \text{ resolve } C_4 \text{ and } C_3 \text{ on } Q \end{aligned}$$

The empty clause $\square$ is derived $\Rightarrow$ the original entailment holds. $\square$

# Resolution Refutation: Example [2]

**Theorem 6** (Completeness of Resolution): Resolution is *refutation-complete*: a set of clauses $S$ is unsatisfiable if and only if the empty clause $\square$ can be derived from $S$ by resolution.

**Why this matters for SAT solving:** Every CDCL solver is *implicitly* building a resolution proof. When a solver reports UNSAT, its learned clauses form a resolution refutation. This is how SAT solvers produce *verifiable certificates* of unsatisfiability — a crucial property for trustworthy verification.

# Proof Systems: Comparison

| System | Human-friendly | Automateable | For SAT | Certify UNSAT |
|---|---|---|---|---|
| Truth tables | Medium | Trivial | No | No |
| Natural Deduction | High | Low | No | No |
| Semantic Tableaux | Medium | Medium | Partial | Yes |
| Resolution | Low | High | Yes | Yes |

Progression from left to right mirrors the course: human methods $\Rightarrow$ machine methods.

SAT solvers are *resolution engines* augmented with heuristics (VSIDS, restarts, phase saving).

# First-Order Logic

# Why First-Order Logic?

Propositional logic handles Boolean constraints well, and SAT solvers are remarkably efficient.
But *specifications* in formal methods require quantification:

```
def binary_search(arr, key):
    # Precondition: arr is sorted
    # Postcondition: returns index of key, or -1
```

The postcondition says: *"For all inputs* $(\mathrm{arr}, \mathrm{key})$ *where* `arr` *is sorted, if the function returns* $i \geq 0$*, then* `arr[i] == key`*; if it returns* $-1$*, then* `key` *is not in* `arr`*."*

In FOL: $\forall\ \mathrm{arr}, \mathrm{key}.\ \mathrm{Sorted}(\mathrm{arr}) \rightarrow (\mathrm{result} \geq 0 \rightarrow \mathrm{arr}[\mathrm{result}] = \mathrm{key}) \wedge (\mathrm{result} = -1 \rightarrow \forall i.\ \mathrm{arr}[i] \neq \mathrm{key})$

**Why PL fails:** Specifications quantify over *unbounded* or *infinite* domains:

- "All array indices are in bounds" $\longrightarrow$ quantifies over all integers $i$ in range
- "No null pointer dereferences" $\longrightarrow$ quantifies over all pointers in the heap
- "The loop preserves the invariant" $\longrightarrow$ quantifies over all iterations

PL would need infinitely many propositions. FOL makes this expressible and *checkable*.

# Why First-Order Logic? [2]

FOL adds three key ingredients:

- **Variables** ranging over objects in a domain ($x, y, z, \ldots$)
- **Quantifiers** ($\forall, \exists$) for generalization
- **Functions** and **predicates** giving structure to the domain

*Example*: "Every prime greater than 2 is odd":

$$\forall p.\, (\mathrm{Prime}(p) \land p > 2) \to \mathrm{Odd}(p)$$

One formula replaces infinitely many propositional checks.

# FOL at a Glance

A first-order *signature* $\Sigma = \langle \mathcal{F}, \mathcal{R} \rangle$ declares function symbols (including constants) and relation symbols.

- *Terms* are built from variables and functions.
- *Formulas* combine terms via predicates, connectives, and quantifiers $\forall, \exists$.

**Syntax example** (arithmetic):

- Signature: $\Sigma = \langle \{0, S, +, \times\}, \{<, =\} \rangle$
- Formula: $\forall x.\, (x = 0 \lor \exists y.\, S(y) = x)$
- Meaning: "every natural number is either 0 or a successor"

**Verification example** (arrays):

- Signature: array operations `read`, `write`, `len`
- Formula: $\forall a, i, v.\, 0 \leq i < \mathrm{len}(a) \to \mathrm{read}(\mathrm{write}(a, i, v), i) = v$
- Meaning: "writing then reading gives back the value"

A *structure* (model) $\mathfrak{A}$ gives meaning to the symbols: a domain $A$, concrete functions, concrete relations. The *same* formula can be true in one structure and false in another — *validity* means truth in *all* structures.

- PL: $2^n$ truth assignments (finite) $\Rightarrow$ SAT is decidable (NP-complete)
- FOL: structures can have *infinite* domains $\Rightarrow$ validity is *undecidable* (Church–Turing, 1936)
- **FM response:** Restrict to *decidable fragments* $\Rightarrow$ SMT theories (linear arithmetic, arrays, *etc.*)

# FOL: Key Concepts Preview

A variable $x$ in $\forall x.\, \varphi$ is *bound*; a variable not in scope of any quantifier is *free*. A formula with no free variables is a *sentence* (has a definite truth value in each structure).

| Concept | Meaning | Example |
|---|---|---|
| Free variables | Unquantified, act as parameters | $x + y > 0$ has free $x, y$ |
| Bound variables | Under $\forall$ or $\exists$ scope | $\forall x.\, x + y > 0$ binds $x$, $y$ free |
| Sentence | No free vars, definite truth value | $\forall x. \exists y.\, y > x$ (true in $\mathbb{Z}$) |
| Structure/Model | Domain + interpretation of symbols | $\mathbb{N}, \mathbb{Z}, \mathbb{R}$ with standard $+, \times, <$ |
| Validity | True in *all* structures | $\forall x.\, x = x$ (valid) |
| Satisfiability | True in *some* structure | $\exists x.\, x \cdot x = 2$ (sat in $\mathbb{R}$, unsat in $\mathbb{Q}$) |

# Why FOL Matters: From Theory to Verification

**Theoretical significance:**

- Gödel completeness (1930): syntactic provability = semantic truth
- Church–Turing (1936): decision problem undecidable
- Compactness: infinite theories have finite character
- Incompleteness (1931): arithmetic has true unprovable statements

**Practical impact on FM:**

- *Dafny* specifications: `requires`, `ensures`, `invariant` are FOL formulas
- *SMT solvers* (Z3, CVC5): decide satisfiability in restricted FOL theories
- *Separation logic*: FOL + heap reasoning for pointer programs
- *Temporal logic*: FOL + time for reactive systems

# Why FOL Matters: From Theory to Verification [2]

*Example (Real verification scenario)*: Dafny method specification:

```
method Find(a: array<int>, key: int) returns (index: int)
  ensures index >= 0 ==> 0 <= index < a.Length && a[index] == key
  ensures index == -1 ==> forall i :: 0 <= i < a.Length ==> a[i] != key
```

The `forall i :: ...` is FOL quantification over integers. Z3 (the SMT solver behind Dafny) checks this by:

1. Translating to many-sorted FOL (separate sorts for `int`, `array<int>`)
2. Applying decision procedures for linear integer arithmetic + array theory
3. Returning SAT (code correct) or UNSAT + counterexample (bug found)

# Automated Reasoning

# From Logic to SAT Solving

The thread from logic to automated reasoning:

1. **Formulas** express constraints about system behavior and specifications.
2. **Normal forms** (especially CNF) provide the *input format* for SAT solvers.
3. **Equisatisfiability** (Tseitin) ensures compact, polynomial-size encodings.
4. **Resolution** is the *theoretical backbone* of DPLL and CDCL solvers.
5. **FOL and theories** motivate SMT solvers — SAT + specialized theory reasoning.
6. **Soundness and completeness** guarantee correctness of the entire pipeline.

**Theorem 7** (Cook–Levin Theorem (1971)): The Boolean satisfiability problem (SAT) is **NP-complete**: every problem in NP can be reduced to SAT in polynomial time.

SAT is the *universal search problem*: if you can verify a solution efficiently, you can encode the search as a SAT instance. Modern CDCL solvers routinely handle formulas with *millions* of variables.

**Next:** SAT encodings, DPLL, CDCL, then FOL theories and SMT.

# Key Takeaways

**Propositional Logic:**

- Syntax (WFFs) vs Semantics (truth values)
- SAT $\iff$ VALID duality via negation
- Equivalence laws $\Rightarrow$ NNF $\Rightarrow$ CNF
- Tseitin: polynomial equisatisfiable CNF

**Proof Systems:**

- Natural deduction: intro/elim symmetry
- Fitch notation for human-readable proofs
- Refutation: assume $\neg\alpha$, derive $\bot$
- Resolution: single rule on clausal form
- Cook–Levin: SAT is NP-complete

**First-Order Logic (taste):**

- Quantifiers ($\forall$, $\exists$) + predicates + functions
- Structures give meaning to symbols
- Same formula: true in one model, false in another
- Full treatment in Weeks 4–5

**What's next:**

- Week 3: SAT encodings, DPLL, CDCL
- Weeks 4–5: FOL deep dive + metatheorems
- Week 6: SMT = decidable fragments of FOL
- Weeks 9–12: Dafny (verification in practice)

**The pipeline we build in this course:**
Specification $\Rightarrow$ logical formula $\Rightarrow$ normal form $\Rightarrow$ solver (SAT/SMT) $\Rightarrow$ verdict.

# Exercises: Propositional Logic

1. Show that $\{\rightarrow, \bot\}$ is a functionally complete set of connectives.
   *Hint*: Express $\neg p$ and $p \wedge q$ using only $\rightarrow$ and $\bot$.

2. Convert the formula $(P \rightarrow Q) \rightarrow R$ to:
   - NNF
   - CNF (using the distributive law)
   - Clausal form (using the Tseitin transformation)

3. For a chain of $n$ biconditionals $p_1 \iff p_2 \iff \ldots \iff p_{n+1}$:
   - How many clauses does the *equivalent* CNF have?
   - How many clauses does the *Tseitin* encoding produce? Explain the asymptotic difference.

4. Show that the satisfiability problem for DNF formulas is solvable in polynomial time.

5. $\star$ Show that *any* propositional proof system has a *tautology* whose shortest proof is exponential in the formula size (assuming NP $\neq$ co-NP). What does this imply about the possibility of efficient general-purpose provers?

# Exercises: Proof Systems

1. Prove the following using natural deduction (Fitch notation):
   - $A \to B, \neg B \vdash \neg A$ *(modus tollens)*
   - $\vdash (A \to B) \vee (B \to A)$
   - $P \to \neg P \vdash \neg P$
   - $\neg(A \wedge B) \vdash \neg A \vee \neg B$ *(De Morgan, requires classical reasoning)*
2. Construct a semantic tableau to test the validity of: $P \to (Q \to R) \vDash (P \to Q) \to (P \to R)$
3. Use resolution refutation to show that $\{P \vee Q, \neg P \vee R, \neg Q \vee R\} \vDash R$.
4. $\star$ Prove that resolution is *not* polynomially bounded: the *pigeonhole principle* $\mathrm{PHP}_n^{n+1}$ (in CNF) requires exponentially long resolution proofs. *(State the formulation and explain why this matters for SAT solving.)*